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Air quality indices: A review of methods to interpret air quality status

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ABSTRACT

Urban air pollution is rapidly increasing in Indian cities. Different methods have been designed to calculate the air quality index in form of mathematical formula. Many studies have shown association between exposure of the population to air pollutants and its adverse health impact. But the formula designed by CPCB in 2014 is more robust to find out the air quality category. To provide the public, information for air pollution dangers, several governmental and municipal agencies developed air quality indices, as a timely information about the potential changes in air quality. In this paper review of air quality index shows how AQI constitutes as an effective method to estimate air pollution levels and risks and help the government in effective pollution control measures adopted for reduction of emissions from various sources.
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1. Introduction

The monitoring and evaluation of ambient air quality is the first important step in controlling air pollution. Huge amount of monitoring data often does not convey air quality status to the scientific community, policy makers, regulatory and in particular to the general public in a simple and straightforward manner. In light of the health effects of air pollutants, environmental agencies have been using air quality index (AQI) for public information (about air quality) and data interpretation. Air Quality Index (AQI) of a given area, which is also known as Air Pollution Index (API) [1–4] or Pollutant Standards Index (PSI) [5,6], has been developed and disseminated by many agencies in U.S. Canada, Europe, Australia, China, Indonesia, Taiwan, etc. [7,8]. An "Air Quality Index" may be defined as a single number for reporting the air quality with respect to its effects on the human health [3,9]. In most elaborate form, it combines many pollutants concentrations in some mathematical expression to arrive at a single number for air quality. This approach for the evaluation of air quality is based entirely on the comparison of measured concentration of pollutants with National Ambient Air Quality Standards (NAAQS). Segmented linear functions are used for relating the actual air pollution concentrations (of each pollutant) to a normalized number. As pollutants accumulate the people get exposed to unhealthy pollutant concentration levels which exceed certain acceptable levels.

Air pollution and its impact to the nearby habitation are the most challenging issues which have to be addressed on top most priority. Air pollutants are generated from different sources including both natural and manmade activities or anthropogenic in to the atmosphere [10,11]. A number of studies has been carried out and the findings has been implemented worldwide [12–15]. More focused research has been undertaken based on regional differences with respect to different air pollutants, their characteristics and their role in the air pollution [16–20]. The concentrations of air pollutants and its magnitude varies across different areas which depends upon the types of industrial and other activities which at the end leads to various types of health related hazards [21–25]. For a full proofed solution, a comprehensive monitoring should be carried out as per the national and international guidelines [15]. Air quality dispersion modeling has also a great role to predict the impact of different air pollutants to the nearby areas [26–28]. Many researches has been carried out by using the mathematical tool to find out the Air Quality Index (AQI) which is very easy to understand and converts different air pollutants concentrations to a single number of an area [29–31]. On the basis of AQI, an area can be categories as good, satisfactory, moderately polluted, poor, very poor, and severe [32,33]. AQI has been developed and used in many developed countries since last three decades [1,34] based on which the ambient atmosphere can be categorized [27,33]. AQI is used for reporting daily air quality of how clean or polluted the air is and its associated health effects in the area [35]. National Air Quality Index to strengthen air quality information dissemination

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system for larger public awareness and their participation on air quality management have been initiated by the Ministry of Environment and Forests (MoEF) Government of India and Central Pollution Control Board (CPCB). CPCB proposed the breakpoint concentration method for measurement of AQI for individual pollutants is very much robust way to find out the status of air quality. This was also adopted by China and is USEPA concept of break point concentration level which they have adopted since last decade for their development.

Similar type of studies has also been carried out in different areas to find out an overall idea about the air quality status which helps to take immediate corrective measures to control over emissions in the area [19,36]. In spite of the introduction of cleaner technologies in industries, energy production and transport, air pollution remains a major health risk. Thus, here is a need for timely information about simple change in Pollution levels in industrial/urban area and to maintain the desired level of air quality. Air quality indices are used for local and regional air quality management in metro cities of the world [37]. The AQI (Air Quality Index) is a single number which represents the air pollution level caused by various pollutants. It shows us how clean or polluted our air is focuses on health effects [38,39]. Recently air quality index (AQI) is launched by prime minister of India and are given in report of central pollution control board [40]. An analysis of the annual and seasonal trends of air quality index of urban and industrial area [41,42] using multi-instrument approach and air quality indexing [43]. A novel air pollutant index based on the relative risk of daily mortality associated with short - term exposure to common air pollutants [7].

The Air Quality Index (AQI) is an indicator number used to report the hourly or daily air quality in an area or a locality. The primary purpose of AQI is to protect public health, especially the health of sensitive people such as the elderly, children and asthmatics. It is used by regulatory agencies to inform people or the public on how clean or polluted the air in their environment is, and its associated health effect. Pollutant Standard Index is a sub-index of the AQI used to calculate pollution level of each air pollutant. While AQI considers all the pollutants, PSI considers each pollutant [44]. PSI and AQI are important to air quality study because they report the actual concentrations of each air pollutants and their degree of pollution and effects. The quality of air we breathe directly affects our quality of life as study has shown that an average human being requires about 12 kg of air each day [45]. Air quality index is used by the United States Environmental Protection Agency (USEPA) to report daily air quality conditions to the public so that sensitive populations can take precautionary measure. The AQI reporting conditions range from good to hazardous [46].

Although, the use of this approach to some extent helps to maintain a 'desired' environmental quality, it does a little to 'map' periodic degradation in air quality, particularly if the measured values remain below NAAQS. The reason behind this drawback arises from the fact that by providing an upper threshold concentration value in the form of a standard, air quality tends to get categorized either as 'good' or 'bad' depending on whether the standards have been exceeded on not. In reality, however, there are instances where concentration of pollutants become sufficiently high to pose environmental and health problems, but owing to the fact that may not be high enough to cross the threshold value, they are falsely interpreted to represent 'acceptable' air quality. Viewed in this backdrop, the present paper attempts to propose an air quality depreciation index that measures deterioration in air quality (with due weightage to the potential capacity of pollutants to affect bio-physical, health and aesthetic attributes) on an 'absolute' environmental quality scale independent of NAAQS. The index makes use of value function curves to determine air quality, an approach conceptually similar to the environmental evaluation system developed by [47] for assessing environmental impacts related to water resource projects. Many researchers have used this concept for determining the ambient air quality.

2. Concept of air quality index

2.1. Air quality index (NAAQS dependent)

In this method to find the air quality index, equal importance is given to all the pollutants. Using observed and standards value, the quality rating for each pollutant is calculated. The geometric mean of these quality ratings gives the Air Quality Index. Based on this assumption, the air quality index can be derived in the manner outlined as under:

The existing concentrations of pollutants are compared with ambient air quality standards (with the standard being assumed as reference base line for each pollutant) and accordingly the quality rating for a particular pollutant is derived as shown below:

$$Q_i = 10(C_i/S_i)$$

where

Q_i = quality rating for a ith pollutant,
C_i = concentration of ith pollutant
S_i = air quality standard for ith pollutant

$$\text{Air Quality Index (AQI)} = (Q_1 \times Q_2 \times \cdots \times Q_n)^{1/n}$$

where n = number of pollutants considered.

Following the above procedure Air Quality Index (AQI) was calculated for all the monitoring stations on annual basis and rankings were assigned as shown in Table 1. The categorization has been assigned as per guideline provided by Environment Canada (www.ec.gc.ca).

2.2. Air quality depreciation index

The air quality depreciation index, as proposed here, attempts to measure deterioration in air quality on an arbitrary scale that ranges between 0 and −10. An index value of '0'represents most desirable air quality having no depreciation from the best possible air quality with respect to the pollutants under consideration while an index value of −10 represents maximum depreciation or worst air quality. Index values differing from 0 towards −10 represent successive depreciation in air quality from the most desirable. The air quality depreciation index is defined as follows:

$$AQ_{dep} = \sum_{i=1}^{n} ((AQ)_i \times CW_i) - \sum_{i=1}^{n} CW_1 \quad (1)$$

where,

AQ_i = Air quality index value for ith parameter
CW_i = Composite weight for ith parameter
n = Total no. of pollutants considered

The values of the AQ_i are determined from the value function curves. In the value function curves the value of 0 signifies worst air quality and value of 1 represents the best air quality for corresponding pollutant concentration. Typical value function curves for SPM, SO_2, NO_x and (TSP × SO_2) are given in Figs. 1–4, respectively.

Value of CW_i in Eq. (1) is computed using the following expression:

$$CW_i = \frac{TW_i}{\sum_{i=1}^{n} TW_i} \times 10 \quad (2)$$

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Table 1
Air Quality Health Index.

Level of risk	Index reading	At risk population	At general population
Low	1–3	• Enjoy your usual outdoor activities. • Follow your doctor's advice for exercise.	Ideal conditions for outdoor activities: sports, biking or walking.
Moderate	4–6	• If you have heart or breathing problems and experience. Symptoms, consider reducing physical exertion outdoors or rescheduling activities to times when the index is lower. • Follow your doctor's usual advice about managing your condition.	No need to modify your usual outdoor activities.
High	7–10	• Children, the elderly and people with heart or breathing problems should reduce or reschedule physical exertion outdoors to periods when the index is lower, especially if you experience symptoms. • Follow your doctor's usual advice about managing your condition.	Anyone experiencing discomfort such as coughing or throat irritation should consider reducing or rescheduling strenuous outdoor activities to periods when the index is lower.
Very High	Above 10	• Children, the elderly and people with heart or breathing problems should avoid physical exertion outdoors. • Follow your doctor's usual advice about managing your condition.	Everyone should consider reducing or rescheduling strenuous outdoor activities to periods when the index is lower, especially if you experience symptoms.

Source: www.ec.gc.ca.

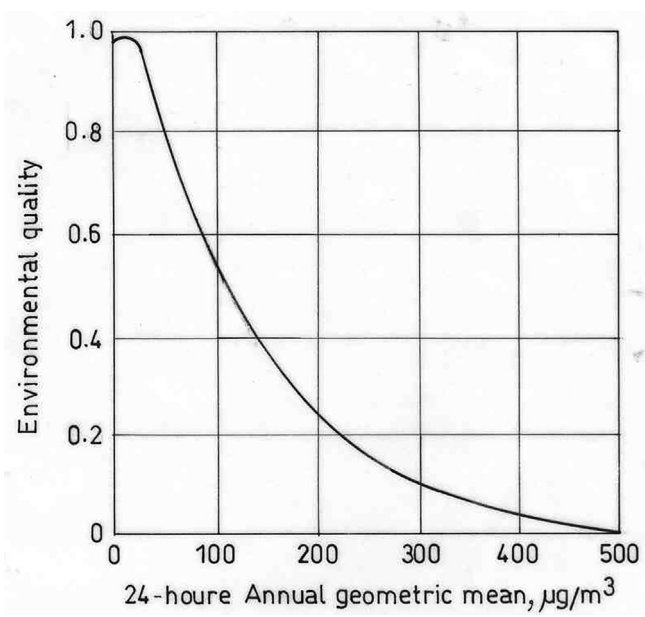

Fig. 1. Value Function Curve for Suspended Particulate Matter (Jain et al., 1977) [48].

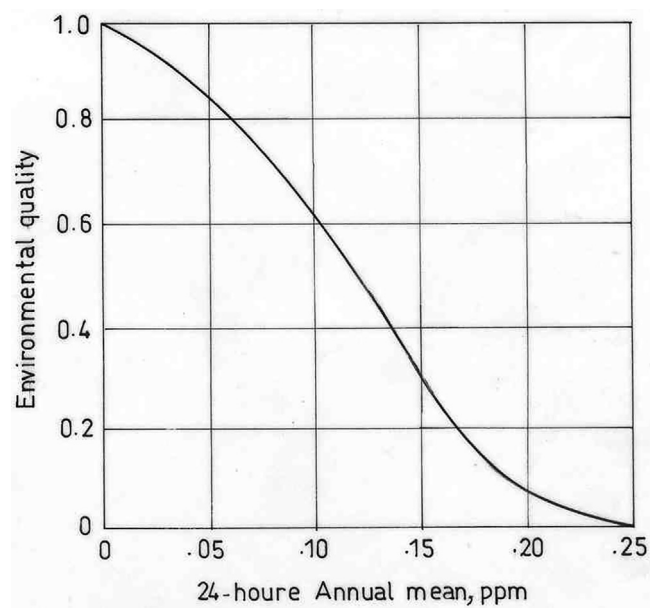

Fig. 2. Value Function Curve for Sulphur Dioxide (Jain et al., 1977) [48].

where,

TW_i = Total weight of ith parameter

$$= AW_i + BPIW_i + HW_i \quad (3)$$

where,

AWi = Aesthetic weight for ith parameter
BPIWi = Bio-Physical Impact Weight for ith parameter
HWi = Health Weight for ith parameter

In computing $TW_{i,}$ an importance weight between 1 and 5 is subjectively assigned to AWi, BPIWi and HWi (i.e., for the ith pollutant) by a team of assessors or experts. Least important assignment is 1 and most important marking is 5. The weights are then aggregated in accordance with Eqs. (2) and (3).

3. Key air pollutants

Air pollution is a worldwide problem and is reported to cause the greatest damage to the health of living thing especially in the Asian countries. Stationary source includes power plants, industrial waste incinerators, the emission of dust from urban construction works and quarries and open burning. Transboundary pollution is defined as the transported air pollution from forest fires from neighbouring countries.

The six key pollutants that released in the atmosphere are particulate matter, ozone, nitrogen dioxide, sulphur dioxide, lead and carbon monoxide. Particular matter categorized by aerodynamic diameter such as particle less than10 μm is known as thoracic particle (PM10), all particle less than 2.5 μm is defined as fine particle, all particle less than 0.1 μm is defined as ultrafine particles (UFP) and the particle between 2.5 μm and 10 μm which is known as coarse particle (PM2.5-PM10). The micro size of particle matter can make it easily enter the respiratory tract and can cause asthma. The main source of this pollutant comes from combustion such as combustion of wood and biomass fuel.

Particulate matter is a type of air pollutants that exists everywhere around us. Particulate matter is a common airborne pollutant, which consists of a complex mixture of liquid, semi-liquid and solid particles which are suspended in the atmosphere [50,51]. It is generated through industrial activities, combustion for energy production, vehicular emissions, fossil fuel combustion

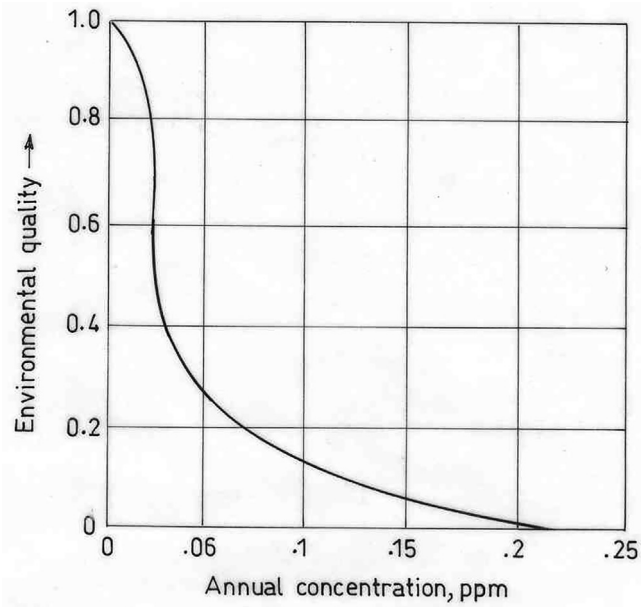

Fig. 3. Value Function Curve for Nitrogen Oxides (Jain et al., 1977) [48].

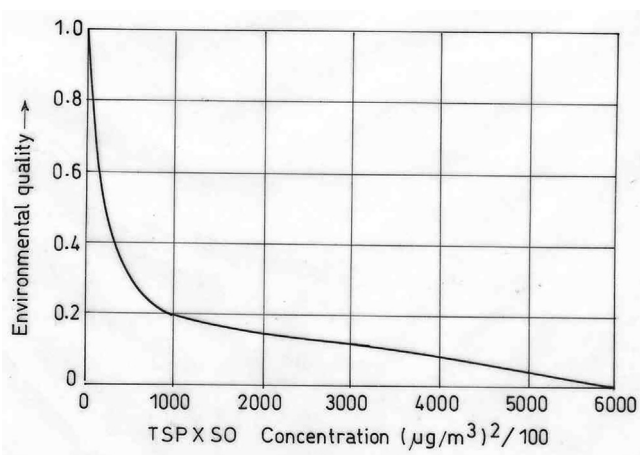

Fig. 4. Value Function Curve for TSP X SO_2 (Luhar & Khanna, 1988) [49].

and range of human activities [50,51]. The sizes of particulate matter used as indicators to describe its health effects are particles with sizes are less than 10 μm in diameter (PM10) and those less than 2.5 μm in diameter (PM2.5). Several health risks have been associated with PM10 and PM2.5 particulate matter. Inhaling particulate matter affects respiratory and cardiovascular systems in both adults and children. Fine particles (such as PM2.5 particulate) has the ability to penetrate deep into the lungs and blood streams when inhaled causing respiratory problems, lung cancer, heart attack and even death [52]. Studies have shown that short term and long-term exposure to particulate matter aggravate cases of asthma, respiratory symptoms among sensitive groups such as young children and the elderly [50,51,53].

Ozone is made up of three oxygen atoms. Ozone is formed in the atmosphere by photochemical reaction in the present of sunlight and pollutant such as oxides of nitrogen and volatile organic compound. It is destroyed by reactions with nitrogen dioxide and it is deposited to the ground [54]. Source of ozone pollution are coming from chemical solvent and combustion product of fuel. Exposure to ozone for a few hours will lead to chest pain, reduces lung function, coughing and asthmatics.

Nitrogen dioxide is very reactive gasses known as nitrogen oxides. It will be produced when fossil fuels such as coal, natural gas are burned. Nitrogen dioxide also emitted by motor vehicle and stationary source such as electric utilities and industrial boilers. Exposure to nitrogen dioxide for will lead to influenza and reduces lung function.

Sulphur dioxide is poisonous gas. It can dissolve with water to form sulphuric acid. The sulphuric acid is the main component of acid rain. Sulphur dioxide is commonly released from the industry section such as a factory that burn high sulphur coal, petroleum refineries and solid waste that contains hazardous and medical waste. Study shows that exposure to acid rain will cause skin cancer, asthma and bronchitis. Lead is an abundant heavy metal in the earth, about 14 parts per million by weight or one part per million by moles. Lead is grey colour, soft and ductile metal. It rarely occurs in pure form in nature. Lead is usually found in ores, mostly with copper, zinc and silver. The most common lead mineral is galena, which is lead sulphide (PbS) and other minerals include lead carbonate and lead sulphate [55]. Lead has become widely distributed in the environment since it was discovered and used by humans for a long time [56]. Lead is an important metal and be used in many industrial especially in the electric battery industry. Lead emissions come from motor vehicle, fossil fuel and manufacturing and mining will cause to premature birth and neurological impairments such as seizures, mental retardation and behavioural disorders. Carbon dioxide is made up from one carbon and two oxygen atoms. Carbon dioxide is released on combustion to the atmosphere. Source of carbon dioxide pollution are mainly come from burning of fossil fuels and production of cement. Exposure to carbon dioxide will effects the range from physiologic (e.g. ventilatory stimulation), to toxic (e.g. cardiac arrhythmias and seizures), anaesthetic (significantly depressed CNS activity), and lethal (severe acidosis and anoxia) [57]. However, the effect of carbon dioxide on the individuals is depending on age, health, occupation, physical activities and lifestyle.

Thus, the air quality index takes into account these six air pollutants.

4. Major air pollutants and their health impacts

Short- and long-term health effects of air pollutants has raised a great concern but also on the ill effects that are often dependent on environmental conditions, dose, and individual susceptibility. Short-term effects are temporary and range from simple discomfort, such as irritation of the eyes, nose, skin, throat, wheezing, coughing and chest tightness, and breathing difficulties, to more serious states, such as asthma, pneumonia, bronchitis, and lung and heart problems. These problems can be aggravated by extended long-term exposure to the pollutants, which is harmful to the neurological, reproductive, and respiratory systems and causes cancer and even, rarely, deaths.

The long-term effects are chronic, lasting for years or the whole life and can even lead to death. Furthermore, the toxicity of several air pollutants may also induce a variety of cancers in the long term [58]. Particulate Matter (PMs), dust, benzene, and O_3 cause serious damage to the respiratory system [59]. Moreover, there is a supplementary risk in case of existing respiratory disease such as asthma [60]. Susceptible people as those with lung disease, old people, and children, who present a higher risk of damage. The major health problems associated with sulfur dioxide emissions in industrialized areas are respiratory irritation, bronchitis, mucus production, and bronchospasm, as it is a sensory irritant and penetrates deep into the lung converted into bisulfite and interacting with sensory receptors, causing bronchoconstriction [60]. Nitrogen dioxide leads

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to respiratory illness and symptoms such as eye, throat, and nose irritation [61].

5. Conclusions

This brief review on air quality indices shows the wide interest or concern for poor air quality problem. But at the same time, it shows lack of a common strategy, which allow to compare the state of the air for cities that follow different directives. The major differences among the indices in the literature are found in the type and number of pollutants, number of index classes (and their associated colours) and related descriptive terms. It was observed that the guidelines are sometimes consistently different from place to place, not only in indicating the pollutants to be monitored, but also in setting the threshold values.

This gives credence to the fact that simple comparisons to NAAQS may not be the best approach to assess air quality. So, the application of the proposed Air Quality Depreciation Index allows for more realistic air quality assessment as compared to interpretive evaluations that revolve around comparing observed concentrations to national ambient air quality standards. The Air Quality Depreciation index can be an invaluable tool to map periodic deterioration in air quality with respect to its potential for environmental damages. It records in a more lucid and meaningful way; even the marginal changes in air quality.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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